



OBJECT ORIENTED SOFTWARE ENGINEERING

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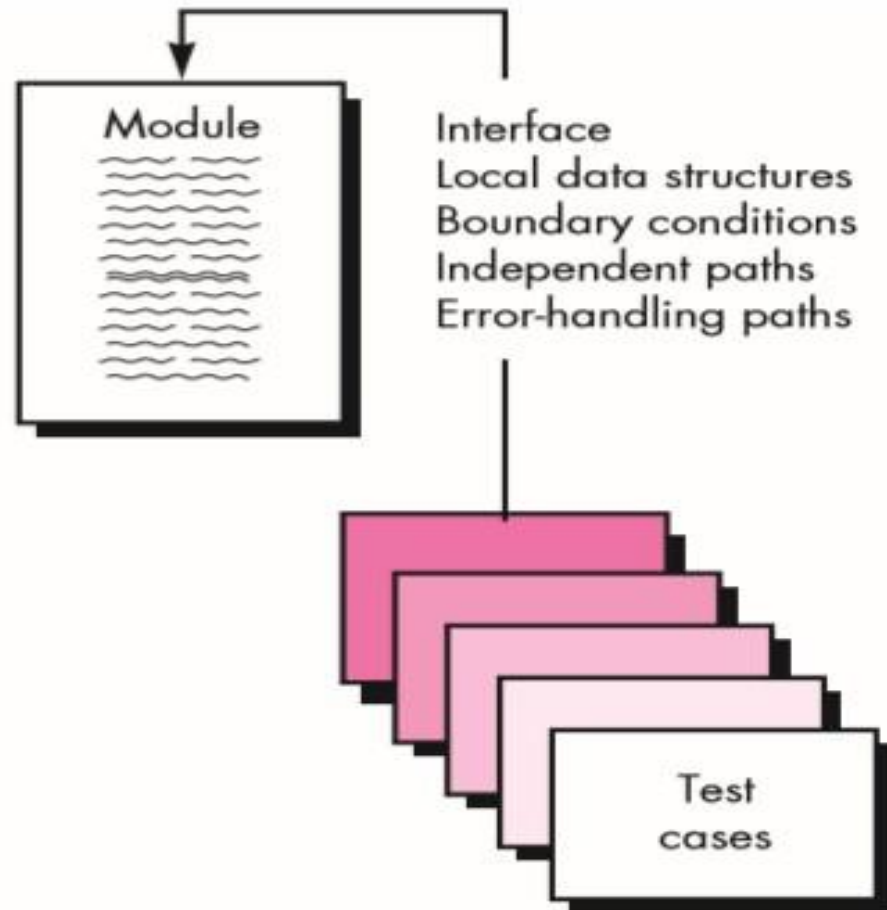
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Unit Testing

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- The first level of testing – unit testing is the **micro level** of testing
- Focuses testing on the **individual unit** or **software module or function** to make sure that they work properly
- Concentrates on the **internal processing logic and data structures**
- Is simplified when a module is designed with **high cohesion**
 - Reduces the number of test cases
 - Allows errors to be more easily predicted and uncovered

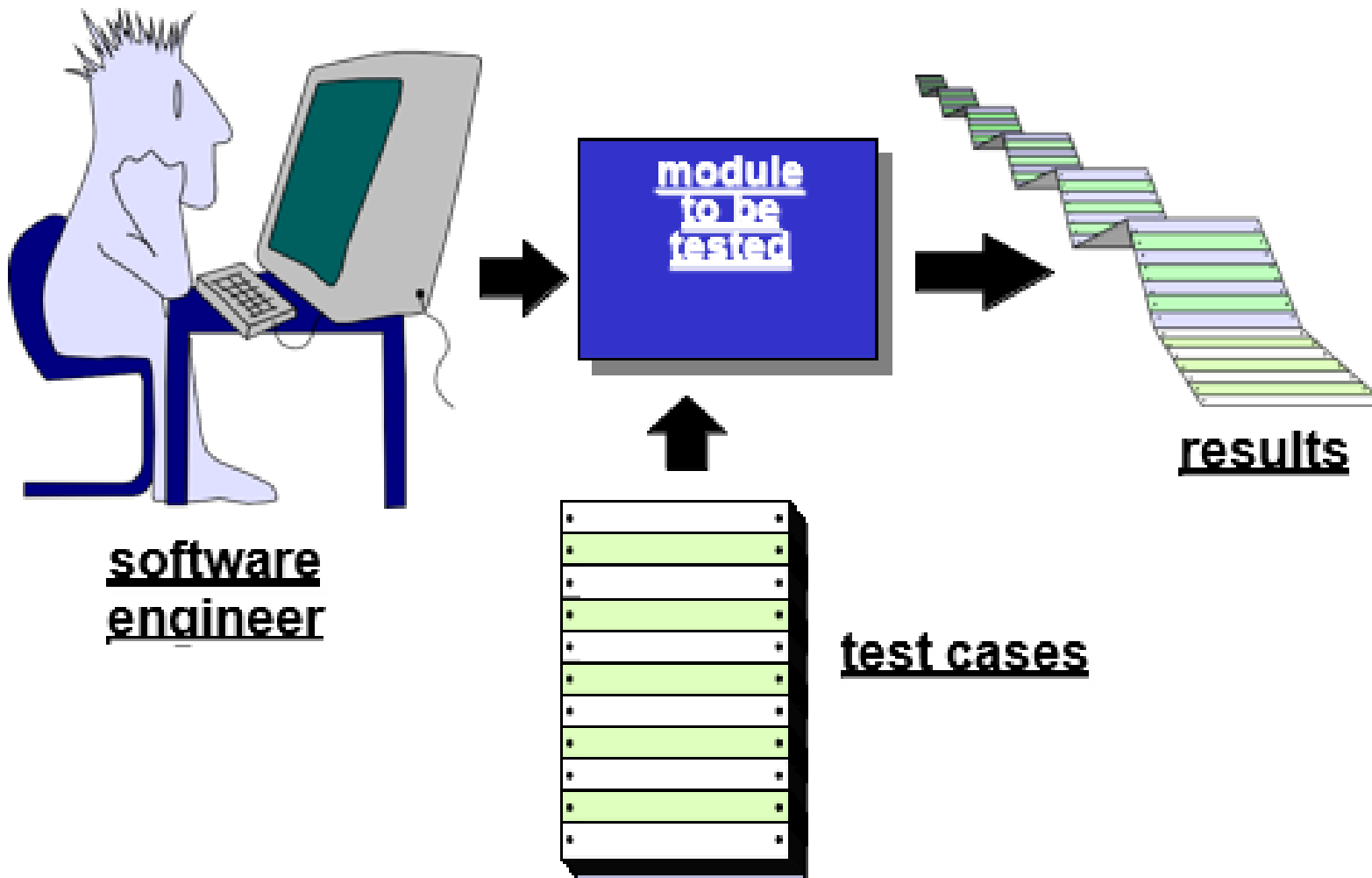


Common Computational Errors in Execution Path

- Misunderstood or incorrect arithmetic precedence
- Mixed mode operations (e.g. int, float, char)
- Incorrect initialization of values Precision inaccuracy and round-off errors
- Incorrect symbolic representation of an expression (int vs. float)

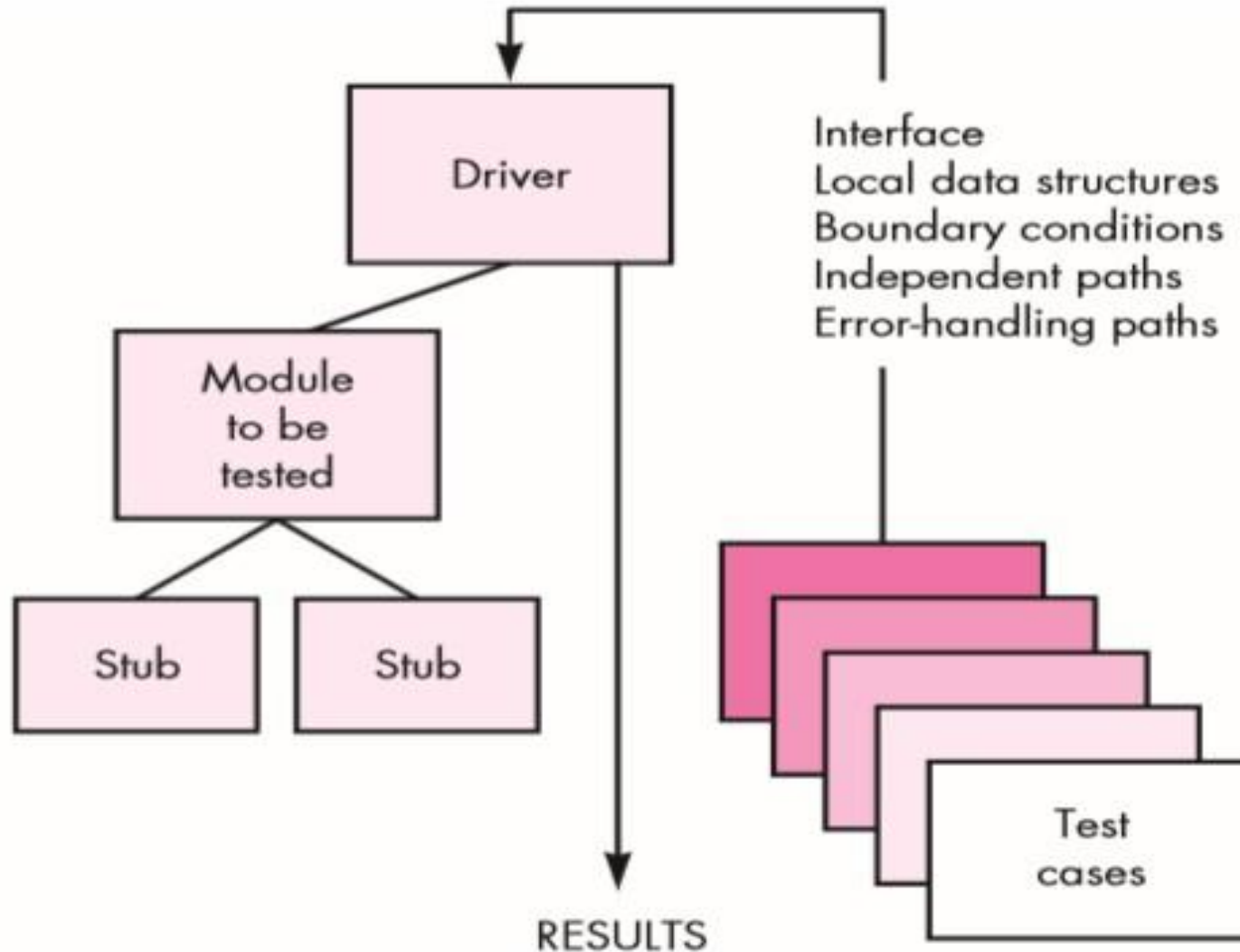
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Unit Testing...



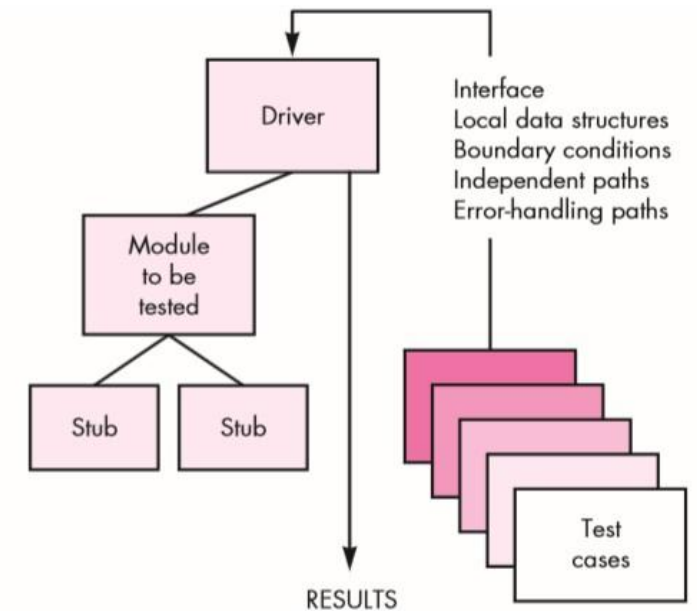
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Unit Test Environment



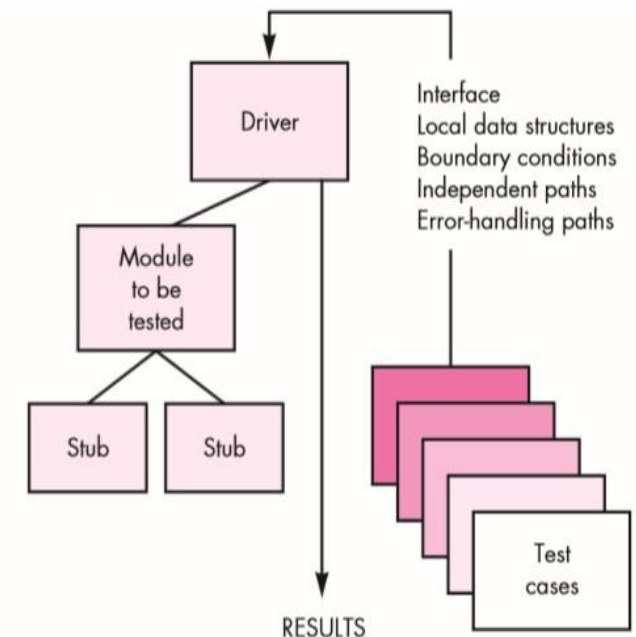
- **Driver**

- A **simple main program** that **accepts** test case data, **passes** such data to the **component being tested**, and **prints the returned results**
- Driver is basically a runtime created **dummy function or method, page or module** which basically input some other module in project.
- Driver is used in **Bottom Up approach** of Integration



- **Stubs**

- Serve to **replace modules** that are subordinate to (called by) the **component to be tested**
- It uses the module's exact interface, may do minimal data manipulation, provides verification of entry, and returns control to the module undergoing testing
- Subordinate modules → that are called by modules to be tested.
- Stub is a **dummy subprogram** that does minimal data manipulation, provides verification of entry and returns the control to the module under testing.



Stubs...

- Stubs are used in **Top Down** approach of **Integration**.
- Drivers and stubs both represent overhead
 - Both must be written but **don't** constitute **part of the installed software product**

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Drivers and Stub for Unit Testing



To be more clear, **consider an example:**

Suppose you have **two pages** of a web application,

- **Page1: Login form** - basically allow the **authenticated and authorized user** to login to application after entering valid credentials.
- **Page2: Dashboard page** – valid user redirected after successful login

- Suppose one developer has finished his work on page1 (login form) and wants to move in initial testing. But page2 (dashboard page) still in development stage.
 - Now the problem arises is page1 cannot land or redirected user to dashboard page since it is still in work progress.
- In this case we create a dummy dashboard page to check the login form's user direction on dashboard page. Such type of dummy is called **Stub**.

- Similarly, to understand what driver is, suppose you have finished page2(dashboard page) and waiting for page1 (login form) to be developed. In this case you create a **dummy to call the dashboard page**. This dummy is called **driver**.

Stub	Driver
Stub is considered as subprogram.	It is a simple main program.
Stub does not accept test case data.	Driver accepts test case data.
It replace the modules of the program into subprograms and are tested by the next driver.	Pass the data to the tested components and print the returned result.



THANK YOU

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